

ARDABIL, Iran, Islamic Republic of

WMO: 407080

Lat: 38.2187N

Lon: 48.3289E

Elev: 4370

StdP: 12.52

Time Zone: 3.50 (IRN)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -0.5	(c) 7.2	(d) -3.6	(e) 5.3	(f) -0.1	(g) 3.5	(h) 7.7	(i) 10.6	(j) 40.0	(k) 43.9	(l) 34.8	(m) 43.4	(n) 2.0	(o) 300	(p) 0.700

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 23.9	(c) 86.2	(d) 63.0	(e) 82.8	(f) 62.4	(g) 80.3	(h) 61.8	(i) 68.5	(j) 78.6	(k) 66.5	(l) 76.7	(m) 64.8	(n) 74.4	(o) 17.4	(p) 70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.2	109.9	73.6	63.3	102.5	71.1	61.5	96.2	69.2	35.7	78.8	33.8	77.0	32.4	74.3	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
32.1	28.7	26.4	-10.3	96.5	5.9	3.6	-14.5	99.0	-17.9	101.1	-21.3	103.1	-25.5	105.7
			-11.8	72.0	7.2	3.5	-16.9	74.5	-21.1	76.5	-25.2	78.5	-30.4	81.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	49.1	29.9	32.9	40.9	47.8	56.9	62.1	65.2	65.4	59.9	52.3	41.5	32.9	(6)
(7)		DBStd	14.74	10.58	9.63	9.68	8.37	5.78	4.48	4.45	4.60	5.47	7.11	8.02	10.14	(7)
(8)		HDD50	2418	622	479	310	136	10	0	0	0	2	58	269	532	(8)
(9)		HDD65	5983	1087	898	746	518	260	109	50	50	169	397	704	995	(9)
(10)		CDD50	2078	0	1	29	69	223	363	472	478	299	128	15	1	(10)
(11)		CDD65	169	0	0	1	1	7	22	58	63	15	2	0	0	(11)
(12)		CDH74	2764	0	0	7	56	215	447	683	934	327	95	0	0	(12)
(13)		CDH80	698	0	0	0	9	39	101	175	263	100	11	0	0	(13)
(14)	Wind	WSAvg	11.2	11.4	10.7	12.2	11.2	10.9	12.2	12.9	12.5	11.2	10.0	9.2	9.9	(14)
(15)	Precipitation	PrecAvg	12.8	1.0	1.0	1.4	1.5	1.9	0.7	0.4	0.4	0.7	1.4	1.5	0.9	(15)
(16)		PrecMax	18.0	1.9	2.3	3.3	3.7	3.5	2.4	1.3	1.1	2.4	4.4	5.0	2.6	(16)
(17)		PrecMin	9.3	0.2	0.2	0.0	0.0	0.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	2.2	0.4	0.6	0.7	0.9	0.7	0.7	0.3	0.3	0.6	1.1	1.2	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.1	59.4	73.4	80.4	84.4	88.1	90.3	91.3	88.7	81.8	67.9	60.7	(19)
MCWB			42.5	44.9	50.2	56.3	59.7	61.5	65.4	64.5	59.8	55.5	49.8	46.3	(20)	
2%		DB	51.5	55.3	67.8	73.4	79.2	82.5	84.4	86.1	82.5	76.8	63.3	55.5	(21)	
		MCWB	40.3	42.4	49.0	54.0	58.4	61.7	64.7	64.5	60.6	54.9	48.0	43.4	(22)	
5%		DB	48.3	51.5	62.7	68.1	75.4	78.9	80.5	82.5	77.3	71.6	59.2	51.7	(23)	
		MCWB	38.9	41.1	46.6	52.1	57.9	61.6	64.1	63.8	60.5	54.4	46.8	41.4	(24)	
10%		DB	44.7	47.8	58.2	64.2	71.5	75.5	77.3	79.1	72.1	66.3	55.3	48.0	(25)	
		MCWB	36.8	39.4	44.9	50.9	56.4	60.4	63.3	62.6	60.1	53.2	45.7	39.5	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.0	46.5	52.1	59.4	65.2	67.8	71.5	72.5	67.3	61.4	53.3	47.7	(27)
MCD B			53.1	57.2	67.4	73.7	75.2	79.2	82.7	83.0	76.9	70.5	62.3	57.1	(28)	
2%		WB	41.4	43.7	49.6	55.9	61.4	65.4	68.5	68.8	64.5	58.3	50.4	44.1	(29)	
		MCD B	49.1	52.8	66.0	69.7	72.7	76.7	78.8	78.6	73.3	66.6	58.6	53.4	(30)	
5%		WB	39.2	41.7	47.5	53.7	59.3	63.5	66.3	66.7	63.0	56.5	48.4	41.9	(31)	
		MCD B	46.6	49.5	61.3	65.9	71.2	74.4	76.7	76.4	71.1	65.5	56.2	49.8	(32)	
10%		WB	37.2	39.6	45.5	51.6	57.5	61.8	64.5	65.0	61.4	54.7	46.5	39.8	(33)	
		MCD B	44.1	47.0	57.1	62.1	68.8	72.3	73.8	74.1	68.5	64.0	54.0	46.7	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	20.5	20.6	22.6	23.9	24.5	23.9	21.9	23.9	21.8	21.8	21.2	20.9	(35)
MCDBR			21.6	25.3	30.7	33.1	32.6	31.9	29.3	32.8	35.1	33.3	30.1	25.8	(36)	
MCWBR			13.3	15.1	16.2	17.0	15.9	16.4	15.6	17.7	18.1	16.5	17.5	16.0	(37)	
5% WB		MCDBR	20.5	23.0	28.6	29.4	29.5	27.0	25.0	26.1	26.3	26.3	23.9	22.9	(38)	
		MCWBR	13.2	14.6	16.0	16.2	16.5	16.1	15.0	15.9	15.5	15.1	14.9	14.7	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.305	0.339	0.389	0.458	0.452	0.444	0.479	0.451	0.413	0.389	0.332	0.303	(40)	
taud		2.353	2.261	2.141	1.986	2.055	2.112	2.030	2.110	2.197	2.218	2.350	2.383	(41)		
Ebn at Noon		276	279	274	261	265	266	256	260	263	257	263	268	(42)		
Edh at Noon		29	36	45	56	53	50	54	49	42	37	29	26	(43)		
(44)	All-Sky Solar Radiation	RadAvg	686	932	1182	1483	1903	2273	2216	2006	1545	1062	736	603	(44)	
(45)		RadStd	63	128	133	167	111	127	115	140	108	115	93	62	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	DBAvg	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A
(47) Regional (9 neighbors)		+0.59	N/A	N/A	+1.75	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page